

api-engine

Skills and Technology Inventory

A platform that converts a natural-language automation goal into a deterministically validated, executable multi-API workflow: it indexes real OpenAPI specifications, retrieves candidate endpoints, has a language model propose a plan, rejects any claim that does not survive seven deterministic checks, and executes the survivor with retries, idempotency and checkpoints.

191 distinct items across 20 subsections. 8 entries mentioned but not used; 5 flagged uncertain.

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Files scanned

pyproject.toml, uv.lock, web/package.json, web/package-lock.json, web/tsconfig.json, web/next.config.mjs, .env, .env.example, web/.env.local.example, .gitignore; 43 Python files under src/, tests/, bench/, docs/ (every import extracted via ast.parse with line numbers); 10 TypeScript/TSX files under web/ (every import statement); web/app/globals.css; bench/queries.json, bench/RESULTS.md; 10 Markdown documents (README.md, CLAUDE.md, architecture.md, techstack.md, conventions.md, env.md, todo.md, EXPLAINED.md, docs/REPORT.md, bench/RESULTS.md).

Not present in this repository (verified by directory walk): Dockerfile, docker-compose.yml, .dockerignore, Makefile, .github/workflows/, .gitlab-ci.yml, .pre-commit-config.yaml, requirements*.txt, Pipfile, environment.yml, poetry.lock, Cargo.toml, go.mod, pom.xml, build.gradle, schema.sql, migrations/, scripts/, bin/, tools/, and any .ipynb notebooks. Verified absent by directory walk.

Programming Languages

Item	Version	Evidence (file, line)
Python 3.13	requires-python >=3.12	pyproject.toml:7; 43 .py files under src/, tests/, bench/, docs/
TypeScript	5.7.3	web/package.json devDependencies; web/tsconfig.json; 10 .ts/.tsx files
JavaScript (ESM)	—	web/next.config.mjs (ES module config)

Query, Markup and Hardware Description Languages

Item	Version	Evidence (file, line)
SQL (SQLite dialect)	—	src/engine/search/retrieve.py:120-133 raw SELECT with JOINS; src/engine/db.py:398 PRAGMA table_info
SQLite DDL — virtual tables	—	src/engine/search/index.py:26 CREATE VIRTUAL TABLE ... USING fts5
SQLite PRAGMA tuning	—	src/engine/db.py:370-372 journal_mode=WAL, synchronous=NORMAL, foreign_keys=ON

FTS5 MATCH query syntax	—	src/engine/search/text.py:build_match_expression (quoted OR disjunction)
HTML	—	docs/render_report.py:build_html; this document
CSS	—	web/app/globals.css (theme tokens, print styles in docs/render_report.py)
JSX / TSX	—	web/app/create/page.tsx, web/app/credentials/page.tsx and 5 more
Markdown	—	10 .md files; parsed programmatically at docs/render_report.py:20
TOML	—	pyproject.toml; uv.lock
JSON Schema (draft 2020-12 vocabulary)	—	src/engine/workflow/schema_match.py – type, required, properties, items
JSON Pointer (RFC 6901 subset)	—	src/engine/ingest/parse.py:_lookup – '~1'/'~0' escape handling

Core CS Foundations

Item	Version	Evidence (file, line)
Depth-first cycle detection (white/grey/black)	—	src/engine/workflow/compiler.py:215 _has_cycle with 'visiting' set
Topological sort (Kahn's algorithm)	—	src/engine/workflow/compiler.py:_topological_order using indegree counts
Breadth-first reachability	—	src/engine/workflow/compiler.py:_reachable_from
Reverse-graph ancestor traversal	—	src/engine/workflow/compiler.py:_ancestors (approval-gate check)
Recursive tree walk with cycle + budget guards	—	src/engine/ingest/parse.py:resolve_refs
Partial selection (quickselect) for top-k	—	src/engine/search/embeddings.py:87 np.argpartition
Directed acyclic graph as the workflow model	—	src/engine/workflow/dag.py – nodes + edges + execution_order
Content-addressed identifiers (SHA-1 digest)	—	src/engine/ids.py:29 hashlib.sha1(...).hexdigest()[:12]
Dotted-path traversal over nested structures	—	src/engine/execution/executor.py:_read_path; schema_match.py:resolve_path

Numerical and Data Libraries

Item	Version	Evidence (file, line)
NumPy	2.5.2	src/engine/search/embeddings.py:21; matrix ops at :86-93
L2 normalisation	—	src/engine/search/embeddings.py:93 np.linalg.norm(axis=1, keepdims=True)
Vectorised cosine via matrix product	—	src/engine/search/embeddings.py:86 self.vectors @ query_vector

Compressed array serialisation (.npz)	—	src/engine/search/embeddings.py np.savez_compressed / np.savez
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Machine Learning and Deep Learning Frameworks

Item	Version	Evidence (file, line)
fastembed	0.8.0	pyproject.toml dependencies; src/engine/search/embeddings.py:165 TextEmbedding
ONNX Runtime	1.29.0	uv.lock (transitive via fastembed); inference backend for embeddings
HuggingFace Hub	1.28.0	uv.lock (transitive) – model download at first use
tokenizers	0.23.1	uv.lock (transitive via fastembed)

NLP and Computer Vision Libraries

Item	Version	Evidence (file, line)
Porter stemmer (via SQLite FTS5)	—	src/engine/search/index.py:29 tokenize="porter unicode61"
py-rust-stemmers	0.1.8	uv.lock (transitive via fastembed)
Custom camelCase splitter	—	src/engine/search/text.py:_CAMEL_BOU NDARY regex; split_camel()
Custom path tokeniser	—	src/engine/search/text.py:tokenize_p ath – '/chat.postMessage' -> 'chat post message'
Stopword filtering	—	src/engine/search/text.py:STOPWORDS frozenset
Synonym / query expansion	—	src/engine/search/text.py:ACTION_SYN ONYMS, expand_tokens()
Pillow	12.3.0	uv.lock (transitive) – used indirectly by pypdfium2 rasterisation in docs/render_report.py:172

LLM and Agentic Frameworks

Item	Version	Evidence (file, line)
Hand-rolled LLM client abstraction	—	src/engine/agent/llm.py – Protocol LLMClient.complete_json
Intent extraction to a typed schema	—	src/engine/agent/intent.py:parse_int ent -> Requirement
Plan generation constrained to a candidate allowlist	—	src/engine/agent/planner.py:_fabrica ted_id_issues
Bounded self-repair loop	—	src/engine/agent/planner.py:plan_wor kflow (max_repairs, default 3)
Explicit model refusal path	—	src/engine/agent/planner.py – 'cannot_satisfy' handling
Token accounting per plan	—	src/engine/agent/llm.py:Usage(prompt _tokens, output_tokens, calls)
Thinking-budget control	—	src/engine/agent/llm.py:128 generationConfig.thinkingConfig.thin kingBudget

Stub/replay client for deterministic tests	—	src/engine/agent/llm.py:StubClient
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AI Platforms, Model Providers and Coding Assistants

Item	Version	Evidence (file, line)
Google Gemini API (generateContent, v1beta)	gemini-3.6-flash	src/engine/agent/llm.py:GeminiClient ; .env LLM_MODEL
Google Generative Language embeddings API	gemini-embedding-001	.env.example EMBEDDING_MODEL comment; src/engine/search/embeddings.py:_embed_gemini (batchEmbedContents)
OpenAI Chat Completions (JSON mode)	—	src/engine/agent/llm.py:OpenAIClient – response_format json_object
OpenAI Embeddings API	—	src/engine/search/embeddings.py:_embed_openai
BAAI/bge-small-en-v1.5 (local, 384-dim)	384 dims	.env EMBEDDING_MODEL / EMBEDDING_DIMENSIONS; src/engine/search/embeddings.py
Google OAuth 2.0 endpoints	—	src/engine/execution/oauth.py – accounts.google.com/o/oauth2/v2/auth , oauth2.googleapis.com/token
APIs.guru OpenAPI directory	—	src/engine/config.py APIS_GURU_LIST_URL; src/engine/ingest/fetch.py:load_directory

Vector Stores and Retrieval

Item	Version	Evidence (file, line)
SQLite FTS5 full-text index (BM25)	—	src/engine/search/index.py:26-30; bm25() call at retrieve.py:122
Per-column BM25 weighting	—	src/engine/search/retrieve.py:30 BM25_WEIGHTS = (0.0, 10.0, 8.0, 6.0, 1.5, 3.0, 2.0)
Flat (brute-force) vector store	—	src/engine/search/embeddings.py:VectorStore – no ANN index, exact search
Reciprocal Rank Fusion	k = 60	src/engine/search/retrieve.py:204 reciprocal_rank_fusion; RRF_K in .env.example
Hybrid retrieval with graceful lexical fallback	—	src/engine/search/retrieve.py:hybrid_search (falls back on EmbeddingError)
Over-fetch then post-filter	—	src/engine/search/retrieve.py:180 overfetch = limit * 5
Incremental / resumable index build	—	src/engine/search/embeddings.py:_load_checkpoint, build_embeddings(resume=True)

ML and AI Concepts

Item	Version	Evidence (file, line)
Dense text embeddings	384-dim	src/engine/search/embeddings.py
Cosine similarity ranking	—	src/engine/search/embeddings.py:86 (unit vectors, dot product)

Lexical vs semantic vs hybrid ablation	—	bench/run_bench.py --modes bm25,vector,hybrid; docs/REPORT.md §7.1
Vocabulary-mismatch failure analysis	—	docs/REPORT.md §5.4; bench/RESULTS.md known-weaknesses section
Prompt-level hallucination containment	—	src/engine/agent/planner.py SYSTEM_PROMPT ABSOLUTE RULES
Structured output with strict schema rejection	—	src/engine/agent/intent.py ConfigDict(extra='forbid')
Adaptive batch sizing under rate limits	—	src/engine/search/embeddings.py (halve on RateLimited, recover after 10 clean batches)

Statistics and Mathematics

Item	Version	Evidence (file, line)
Precision@K	K=5	bench/run_bench.py:41 precision_at_k
Recall@K	K=20	bench/run_bench.py:53 recall_at_k
Mean Reciprocal Rank (MRR)	—	bench/run_bench.py:60 reciprocal_rank + ModeReport.mrr
nDCG@10 (binary relevance, log2 discount)	K=10	bench/run_bench.py:73 ndcg_at_k
Latency percentiles (p50 / p95)	—	bench/run_bench.py:90 percentile (ceil-rank method)
Arithmetic mean aggregation	—	bench/run_bench.py:120 statistics.fmean
Inverse document frequency weighting	—	inherent to FTS5 bm25(); reasoning in docs/REPORT.md §1.6
Exponential backoff with multiplicative jitter	—	src/engine/execution/executor.py:_backoff_seconds – 0.5 + random.random()

Backend, API and Web Frameworks

Item	Version	Evidence (file, line)
FastAPI	0.141.1	src/engine/api/main.py:16; 26 routes
Starlette	1.6.0	uv.lock (transitive under FastAPI)
Uvicorn (with [standard] extras)	0.52.4	pyproject.toml; run via `uvicorn engine.api.main:app`
Pydantic v2	2.13.4	src/engine/workflow/dag.py:18; src/engine/api/main.py:17
pydantic-settings	2.15.0	src/engine/config.py:12 BaseSettings, SettingsConfigDict
ASGI lifespan context manager	—	src/engine/api/main.py:46 @asynccontextmanager lifespan
FastAPI dependency injection	—	src/engine/api/main.py:68 SessionDep = Annotated[Session, Depends(get_session)]
httpx (sync client, streaming, mounted transports)	0.28.1	src/engine/ingest/fetch.py client.stream; src/engine/agent/llm.py

Typer CLI	0.27.1	src/engine/cli.py:7; 14 commands incl. a sub-app
anyio	4.14.2	uv.lock (transitive under Starlette/httpx)

Frontend and UI Frameworks

Item	Version	Evidence (file, line)
Next.js (App Router)	16.3.2	web/package.json; web/app/** route structure
React	19.0.0	web/package.json; hooks used across 6 components
React DOM	19.0.0	web/package.json
React hooks: useState, useEffect, useCallback, useMemo	—	web/app/credentials/page.tsx:3
React Suspense boundary	—	web/app/oauth/callback/page.tsx:3 Suspense
Next.js route handlers (BFF proxy)	—	web/app/api/proxy/[...path]/route.ts
Next.js catch-all dynamic routes	—	web/app/api/proxy/[...path]/ directory name
next/navigation useSearchParams	—	web/app/oauth/callback/page.tsx:5
Client/server component split ('use client')	—	web/app/page.tsx:1 and 5 others
CSS custom properties with dark-mode media query	—	web/app/globals.css:1-40 (:root tokens + prefers-color-scheme)
TypeScript path aliases	—	web/tsconfig.json paths: {'@/*': ['./*']}

Databases and Data Stores

Item	Version	Evidence (file, line)
SQLite	—	src/engine/config.py DATABASE_URL default sqlite:///data/engine.db
SQLAlchemy 2.0 ORM (DeclarativeBase, Mapped)	2.0.52	src/engine/db.py:28
16 ORM models	—	src/engine/db.py – Provider, Api, Endpoint, Parameter, SchemaRecord, AuthScheme, Credential, CredentialAudit, Workflow, WorkflowNodeRow, CompiledIR, Execution, NodeResult, ExecutionEvent, Job, IngestRun
Composite primary keys	—	src/engine/db.py NodeResult (execution_id, node_id)
JSON columns	—	src/engine/db.py – schemas, tags, definition, ir, payload_metadata
LargeBinary columns for ciphertext	—	src/engine/db.py Credential.ciphertext, refresh_ciphertext
SQLAlchemy event listener	—	src/engine/db.py @event.listens_for(_engine, 'connect') for PRAGMAs
Idempotent upsert via Session.merge	—	src/engine/ingest/run.py:_write_api

Hand-rolled additive migration	—	src/engine/db.py:_add_missing_columns (PRAGMA table_info + ALTER TABLE ADD COLUMN)
Write-Ahead Logging	—	src/engine/db.py:370 PRAGMA journal_mode=WAL
Conditional UPDATE for optimistic claim	—	src/engine/execution/runner.py:131-141 update().where(lease expired)
Append-only audit tables	—	src/engine/db.py CredentialAudit, ExecutionEvent
greenlet	3.5.5	uv.lock (transitive, required by SQLAlchemy)

Cloud, DevOps and Infrastructure

Item	Version	Evidence (file, line)
uv (package + venv manager)	0.12.5	uv.lock present; pyproject.toml [tool.uv] workflow
hatchling build backend	—	pyproject.toml [build-system] requires = ['hatchling']
Console-script entry point	—	pyproject.toml [project.scripts] engine = 'engine.cli:app'
src/ layout packaging	—	pyproject.toml [tool.hatch.build.targets.wheel] packages = ['src/engine']
npm / package-lock	lockfileVersion 3	web/package-lock.json
Headless Chromium via Playwright	1.62.0	docs/render_report.py:131 sync_playwright
Headless Chrome CLI print-to-pdf	—	used to render this document
POSIX signal handling for graceful shutdown	—	src/engine/execution/worker.py:16 signal.SIGINT/SIGTERM
Thread pool for parallel I/O	—	src/engine/ingest/fetch.py:13 ThreadPoolExecutor(max_workers=8)

Version Control and Developer Tools

Item	Version	Evidence (file, line)
Ruff (lint + format)	0.16.4	pyproject.toml [tool.ruff] line-length 100, select E,F,I,UP,B
pytest	9.1.1	pyproject.toml [tool.pytest.ini_options]; 211 test functions
respx (httpx mock transport)	0.23.1	tests/test_execution.py:14; 33 @respx.mock uses
Git ignore rules	—	.gitignore — .env, data/, node_modules, docs/_preview
Environment template pattern	—	.env.example (32 documented keys); web/.env.local.example
Structured logging	—	logging module used in 15 files; noisy loggers suppressed at cli.py:_configure_logging

Scraping, Automation and Messaging

Item	Version	Evidence (file, line)
Streaming HTTP download with byte cap	8 MB	src/engine/ingest/fetch.py:fetch_spec - client.stream + MAX_SPEC_BYTES
On-disk response cache	—	src/engine/ingest/fetch.py - skips download when cache_name exists
OpenAPI 3.x parsing	—	src/engine/ingest/parse.py:_base_url_v3, _request_schema_v3
Swagger 2.0 parsing and normalisation	—	src/engine/ingest/parse.py:_base_url_v2, _response_schema_v2, in:body handling
Provider deduplication by (provider, normalised title)	—	src/engine/ingest/fetch.py:_dedup_key
Slack Web API (chat.postMessage)	—	tests/test_execution.py slack_node(); corpus row slack_com
GitHub REST API v3	—	tests/conftest.py ep_gh_issue_create; corpus row github_com
Gmail API / Google Calendar API	—	src/engine/execution/oauth.py scopes gmail.readonly, gmail.send, calendar.readonly

Product, Design and Collaboration Tools

Item	Version	Evidence (file, line)
Provider setup guides embedded in the UI	—	web/lib/providers.ts - 12 entries with step lists and token URLs
Theme-aware design tokens	—	web/app/globals.css :root + prefers-color-scheme override

Engineering Techniques and Patterns

Item	Version	Evidence (file, line)
Idempotency keys on state-changing calls	SHA-256, 32 hex	src/engine/execution/executor.py:201 ; header set at :349
Checkpoint-before-acknowledge ordering	—	src/engine/execution/runner.py:run_execution - NodeResult written before status flip
Resume from checkpoint	—	src/engine/execution/runner.py:_completed_nodes
Leased job claiming (visibility timeout)	300 s default	src/engine/execution/runner.py:claim_job; QUEUE_LEASE_SECONDS
At-least-once delivery with redelivery counter	—	src/engine/execution/runner.py Job.deliveries
Dead-letter queue	5 deliveries	src/engine/execution/runner.py:finish_job; QUEUE_MAX_DELIVERIES
Retry with exponential backoff and jitter	3 retries default	src/engine/execution/executor.py:_backoff_seconds; HTTP_MAX_RETRIES
Retry-After header honoured	—	src/engine/execution/executor.py - retry_after parsed from 429 response
Failure taxonomy driving retry policy	7 categories	src/engine/execution/executor.py:34 RETRYABLE frozenset

Hidden-failure detection in 2xx bodies	—	src/engine/execution/executor.py:_body_reports_failure ({'ok': false})
SSRF egress guard	—	src/engine/execution/guards.py – scheme allowlist, blocked hostnames, private-range rejection on every resolved address
Redirects refused rather than followed	—	src/engine/execution/executor.py follow_redirects=False; 3xx treated as provider error
Provider execution allowlist (env + database)	—	src/engine/execution/policy.py:is_allowed
Encryption at rest for credentials (Fernet)	—	src/engine/execution/credentials.py:_cipher; cryptography 50.0.0
Key-version field for rotation	—	src/engine/db.py Credential.key_version; env CREDENTIAL_KEY_VERSION
Secret redaction before logging	—	src/engine/execution/credentials.py:redact; src/engine/agent/llm.py:_redact
Credentials as headers, never query strings	—	src/engine/agent/llm.py GeminiClient x-goog-api-key header (comment explains the leak it prevents)
Payload-metadata-only logging	—	src/engine/execution/executor.py:BuiltRequest.describe (names and sizes, never values)
OAuth 2.0 authorization-code flow	—	src/engine/execution/oauth.py:authorize_url, exchange_code
CSRF state parameter with TTL and single use	600 s	src/engine/execution/oauth.py:issue_state/consume_state, secrets.token_urlsafe(24)
Offline access for refresh tokens	—	src/engine/execution/oauth.py:74 access_type=offline, prompt=consent
Transparent token refresh on expiry	—	src/engine/execution/credentials.py:_refresh_in_place
Compile to an immutable intermediate representation	—	src/engine/workflow/dag.py:CompiledWorkflow; stored in compiled_ir
Insert-only workflow versioning	—	src/engine/workflow/store.py; Workflow.parent_version
Deterministic content-addressed IDs	—	src/engine/ids.py – same natural key always yields same id
Unguessable IDs as an anti-hallucination control	—	src/engine/ids.py module docstring; docs/REPORT.md §3.2
Structured machine-readable validation errors	—	src/engine/workflow/dag.py:ValidationIssue (code, field, expected, actual, hint)
Seven-pass deterministic compiler	—	src/engine/workflow/compiler.py – structural, existence, agreement, parameters, body, compatibility, policy
Dry-run mode (build and check, never dispatch)	—	src/engine/execution/executor.py – dry_run branch returns before send
Graceful degradation with visible warning	—	src/engine/search/retrieve.py:hybrid_search logs and falls back to BM25

Backend-for-frontend proxy	—	web/app/api/proxy/[...path]/route.ts – gateway URL never reaches the browser
Request timeout budget with AbortController	120 s	web/app/api/proxy/[...path]/route.ts TIMEOUT_MS
Shared node budget to bound recursion output	20,000 nodes	src/engine/ingest/parse.py:_Budget, MAX_SCHEMA_NODES
Separate limits for cycles, depth and size	8 / 60 / 20000	src/engine/ingest/parse.py MAX_REF_DEPTH, MAX_STRUCTURAL_DEPTH, MAX_SCHEMA_NODES
Per-spec error isolation during ingestion	—	src/engine/ingest/run.py try/except per file, failures recorded in IngestRun
functools.lru_cache for settings singleton	—	src/engine/config.py:8
Protocol-based dependency inversion	—	src/engine/agent/llm.py:LLMClient Protocol
Context-manager session scope	—	src/engine/db.py:session_scope (commit/rollback/close)

Evaluation and Measurement Practice

Item	Version	Evidence (file, line)
Hand-labelled golden query set	30 queries	bench/queries.json
Labels sourced by SQL pattern, never by the ranker under test	—	bench/queries.json 'method' field states this explicitly
Multi-answer relevance judgements	—	bench/queries.json 'relevance' field – several providers may be correct
Deliberate vocabulary-trap queries	—	bench/queries.json q02, q15, q23 notes
Three-way ablation (bm25 / vector / hybrid)	—	bench/run_bench.py:main --modes
A/B of a single configuration flag	—	bench/RESULTS.md 'Synonym expansion' experiment; SEARCH_EXPAND_SYNONYMS
Recorded negative result, change reverted	—	src/engine/search/text.py comment 'Measured experiment, 2026-08-22'; docs/REPORT.md §7.3
Warm-up call excluded from timing	—	bench/run_bench.py:158 search(..., 'warm up the caches')
Per-query miss reporting	—	bench/run_bench.py:ModeReport.misses
Reproducible results committed with date	—	bench/RESULTS.md header
Property-style invariant test across components	—	tests/test_execution.py TestPathSyntaxMatchesTheValidator – every path the compiler accepts must be readable by the executor
Parametrised test matrices	22 sites	tests/*.py @pytest.mark.parametrize
Autouse fixtures for environment isolation	6 sites	tests/test_execution.py, tests/test_oauth.py
Mocked transport so no test reaches a network	33 sites	tests/*.py @respx.mock
In-memory database fixture	—	tests/conftest.py create_engine('sqlite:///memory:')

Deliberately broken inputs as test cases	—	tests/test_compiler.py – fabricated ids, cycles, type mismatches, unsupported auth
Crash-injection / resume test	—	tests/test_execution.py test_a_completed_node_is_not_run_again_on_resume
Secret-leak assertions	—	tests/test_execution.py test_a_secret_never_appears_in_an_error_message
Visual verification of rendered output	—	docs/render_report.py:verify – rasterises pages, flags all-dark or all-blank

Mentioned but not used

Item	Where it was mentioned, and why it does not count
jsonschema 4.26.0	Declared in pyproject.toml dependencies and installed, but there is no <code>import jsonschema</code> anywhere in src/, tests/, bench/ or docs/. Schema checking is hand-written in src/engine/workflow/schema_match.py.
PyYAML 6.0.3	Declared in pyproject.toml. No <code>import yaml</code> anywhere. Intended for YAML OpenAPI specs; the fetcher only ever requests the JSON variant (src/engine/ingest/fetch.py uses swaggerUrl).
python-dotenv 1.2.3	Declared in pyproject.toml. Never imported directly; .env loading is done by pydantic-settings (src/engine/config.py SettingsConfigDict(env_file=...)).
Pygments 2.21.0	Installed as a dev dependency for syntax highlighting. The markdown call in docs/render_report.py:22 does not enable the codehilite extension, so it is not exercised.
subprocess (stdlib)	Imported at docs/render_report.py:16 but never called — a leftover from an earlier rendering approach.
PostgreSQL, OpenSearch, pgvector, Redis, Docker, Kubernetes, OpenTelemetry, Prometheus, Grafana, Celery, React Flow, Alembic	Named throughout techstack.md, architecture.md and todo.md as the mature-system target. None appear in any manifest, import or config. docs/REPORT.md §4.2 states SQLite replaces the first four.
Cross-encoder reranking	Discussed in docs/REPORT.md Part VI as investigated and not adopted. fastembed ships the models but no code imports TextCrossEncoder.
Webhooks / scheduled triggers	Described in architecture.md and todo.md. No receiver route, no scheduler, no triggers table. Listed as the largest gap in docs/REPORT.md Part IX.

Uncertain

Item	Why uncertain, and what would confirm it
mmh3 5.2.1, loguru 0.7.3, protobuf 7.36.0, tqdm 4.70.0, requests 2.34.2, Pillow 12.3.0	Present in uv.lock as transitive dependencies of fastembed and huggingface-hub. No project file imports them. Listed because they are genuinely installed and executed at runtime inside those libraries, but they are not skills the author exercised directly. Confirm by <code>uv tree</code> .

Google embeddings API (gemini-embedding-001)	The code path exists and was exercised during development (src/engine/search/embeddings.py:_embed_gemini), but .env now sets EMBEDDING_PROVIDER=local, so it is not the active path. Confirm by setting EMBEDDING_PROVIDER=gemini and running `engine embed`.
OpenAI provider paths	Both OpenAIClient and _embed_openai are implemented and unit-testable, but .env sets LLM_PROVIDER=gemini and no OpenAI key is configured, so they have not been executed against the live API. Confirm by setting LLM_PROVIDER=openai with a key.
uv version 0.12.5	Read from the install output during this session, not from a file in the repository. uv.lock does not record the uv version that produced it. Confirm with `uv --version`.
Python 3.13 vs 3.12	pyproject.toml requires >=3.12; the interpreter actually used is 3.13.3. Confirm with `python --version`.

Coverage note

Everything listed above was read directly. Python imports were extracted programmatically with ast.parse rather than by grep, so relative imports and multi-name `from x import a, b` statements are captured exactly with their line numbers.

Not read: the contents of .venv/ and web/node_modules/ (installed third-party source, not project code); data/ (ingested OpenAPI specifications, the SQLite database and the embeddings matrix — data artefacts, not code); .next/ build output; and the value of any secret in .env, which was deliberately not opened beyond reading key names.

uv.lock contains 149 locked packages in total. Only those directly imported by project code, or notable transitive runtime dependencies, are listed; the remainder are ordinary transitive infrastructure.